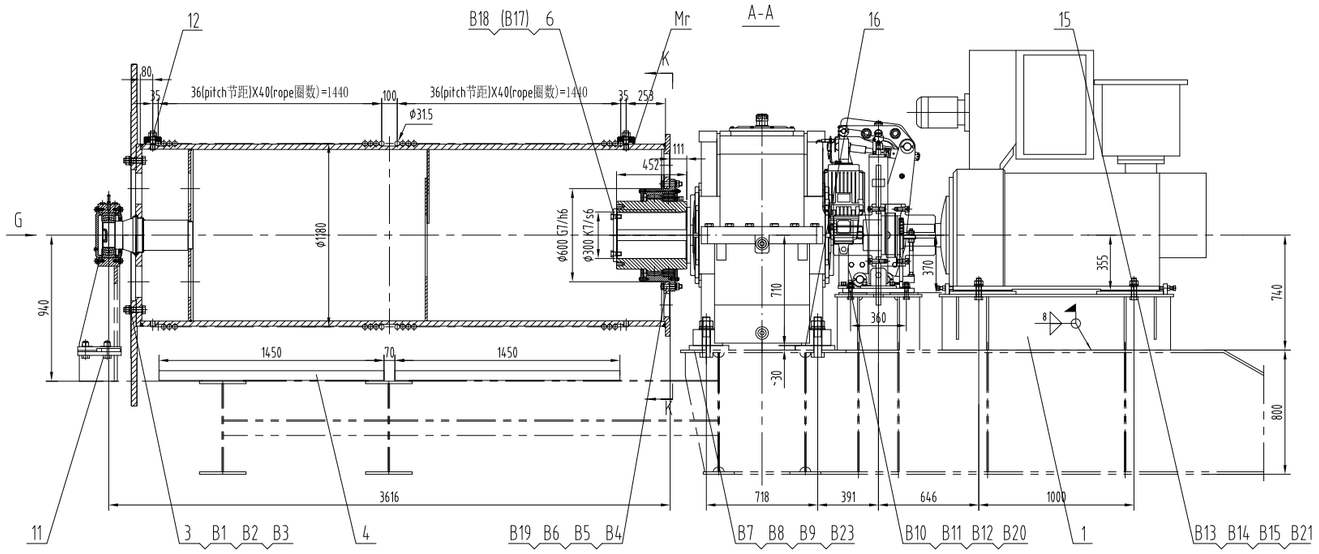
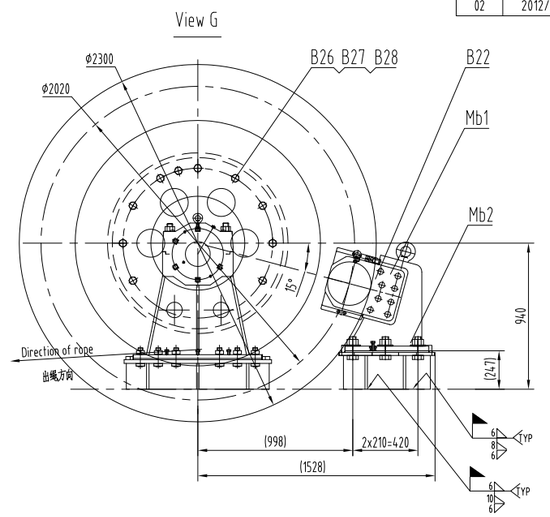
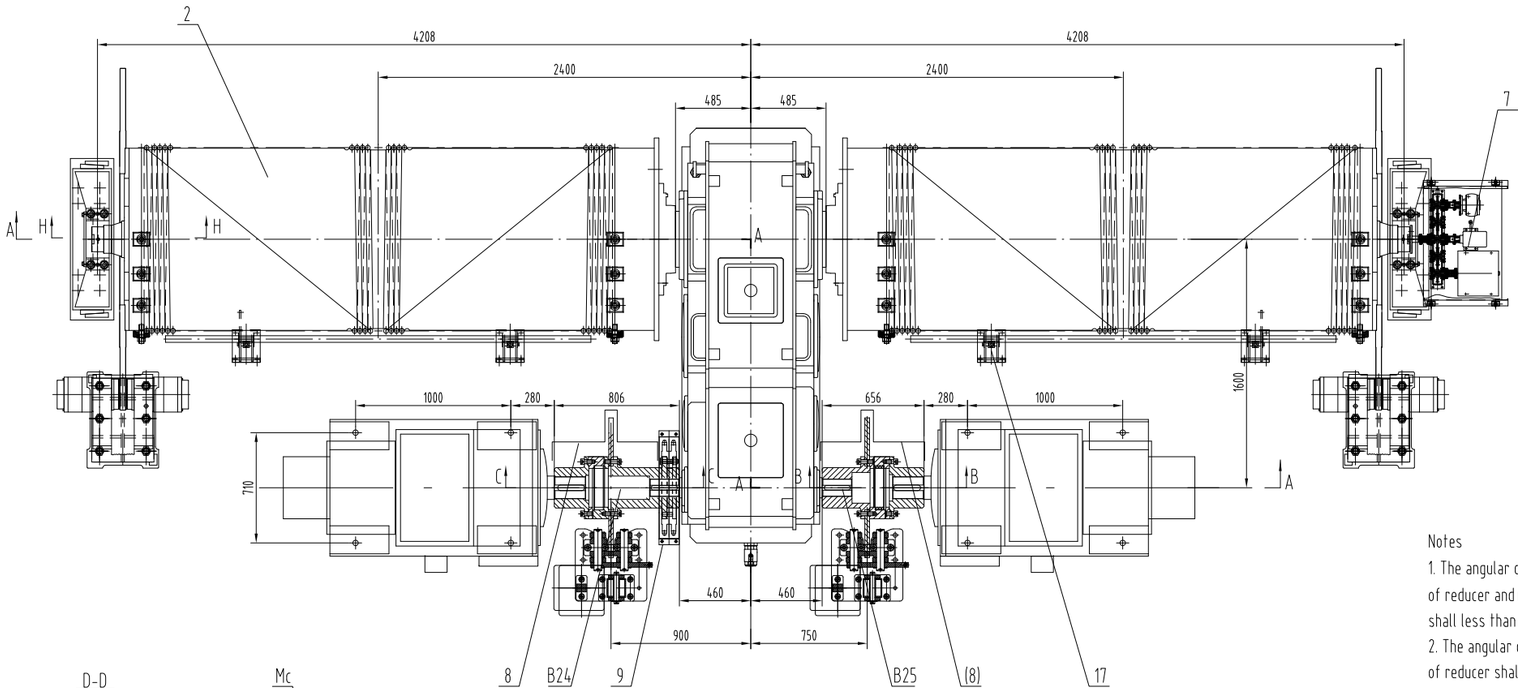
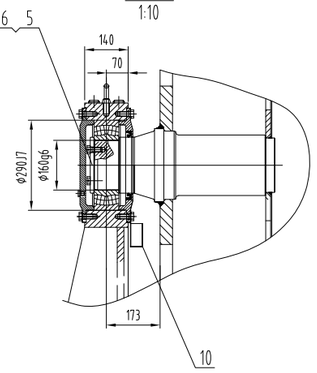
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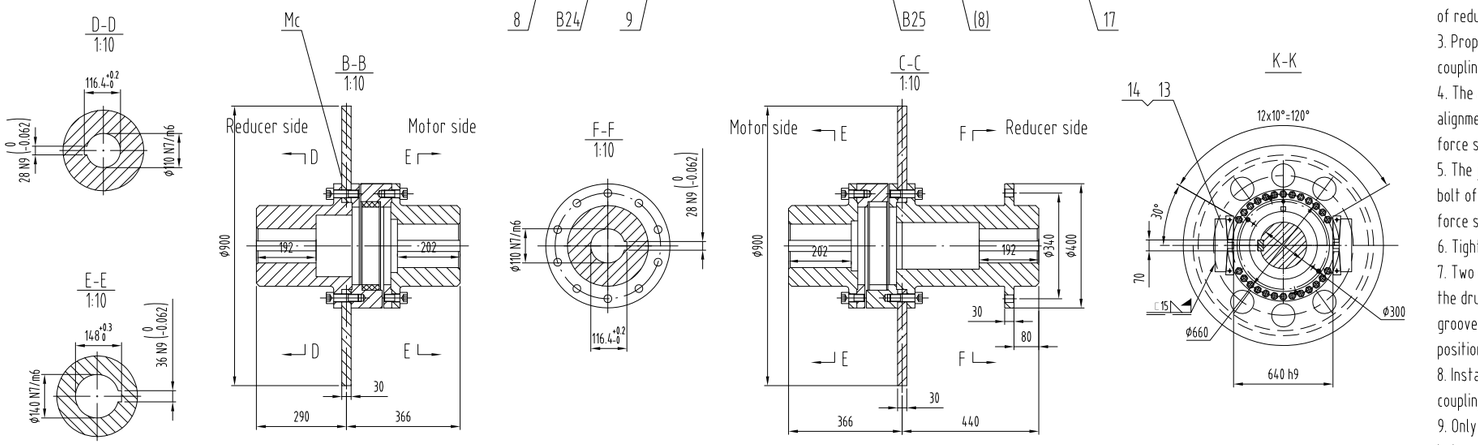








MAIN DATA 主参数		
Load capacity 起升载荷	Under spreader 吊具下	twin:65 T
	Under cargo beam 吊钩横梁下	75T
Hoisting speed 起升速度	Full load 满载	75 m/min
	Empty spreader 空载	180 m/min
Lift height 起升高度	Above rail 轨上	45m
	Under rail 轨下	18m
Motor 电动机	Type 型号	(YASKAWA) BDK-1KMU
	Rated Power 额定参数	2x450Kw
	Application Power 应用功率	2x450Kw
	Application speed 应用转速	750/1800 rpm
Reducer 减速箱	Type 型号	(SEW) ML3PSF130E-1600
	Ratio 速比	18.551
High speed brake 高速制动器	Type 型号	(ZPMC) YP41-Ed4500-80-900x30
	Rated torque 额定力矩	23000Nm
	Set torque 设定力矩	23000Nm
Low speed brake 低速制动器	Type 型号	(ZPMC) SB315
	Braking torque 制动力矩	2x254520 Nm
High speed coupling 高速联轴器	Type 型号	(KTR) AFN SB160-900x30
	Rated torque 公称转矩	19200 Nm
	Braking torque 制动力矩	38400 Nm
Low speed coupling 低速联轴器	Type 型号	(ZPMC) DC08B
	Rated torque 公称转矩	315 KNm
Rope (manho) 钢丝绳	Type 型号	IWRC 6xWS(36)-B-φ31.5
	Breaking strength 破断拉力	672 KN
Drum 卷筒	Nominal Diameter 名义直径	φ1180 mm
Emergency Mechanism 应急机构 — Motor 电机	Power 功率	37 kW
	Speed 转速	1480 rpm
Emergency Mechanism 应急机构 — Reducer 减速箱	Type 型号	FH220.31.G1A-00
	Reducer ratio 减速箱速比	30.852
	Chain wheel ratio 链轮速比	17/35



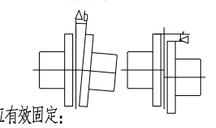
序号 Item No	B2	B5	B8	B11	B14	Mb1	Mb2	Mc	Mr
螺母尺寸D size	M30(10级)(加二硫化钼 (adding MoS_2)	M24(10级)(加二硫化钼 (adding MoS_2)	M48(加二硫化钼 (adding MoS_2)	M24(8级)	M30(8级)	M36(10.0级)	M42-8	M24	M30
拧紧力矩Mo tightening torque	1470Nm	740Nm	4400Nm	710Nm	1400Nm	3500Nm	1700~2100Nm	~1000Nm	900Nm
预紧力Fo tighten force	326.2kN	205.1kN	618.8kN	148.4kN	235.9kN				
扭矩系数k torque coefficient	0.15	0.15	0.15	0.2	0.2				

Notes

1. The angular displacement (△b) between the axes of the high speed shaft of reducer and motor shall less than 2' and the radial misalignment (△r) shall less than 0.05mm. Please refer to figure right.
2. The angular displacement between the axes of drum and low speed shaft of reducer shall less than 10'.
3. Proper amount of grease shall be added into the bearings and gears of coupling.
4. The alignment taper shear blocks shall be well secured after machinery alignment: locating face contract each other sufficiently, the knocking force shall less than 50kg,inside taper blocks be tack welded.
5. The jacking screw shall be well secured after machinery alignment: The bolt of head and locating face contract each other sufficiently, the tighten force shall less than 50kg, lock nuts being tightened.
6. Tighten the bolts to the specified tension force as shown in below table.
7. Two clamped and three unclamped full dead wraps shall be remaining on the drums when the spreader is at its lowest operating position. One empty grooves shall be remaining when the spreader is at its highest operating position.
8. Installing instructions in the manuals of purchased parts such as coupling, brake and etc shall also be conformed.
9. Only epoxy zinc rich primer is permitted to be painted on the fray surface between drum and drum coupling or drum brake disc. The fray surface shall keep clean, dry, and non-polluted before assembly.

技术要求

1. 电机中心线与减速箱中心线的交角(△b)应小于2'，径向相对偏差(△r)应小于0.05mm，见右图



2. 卷筒中心线与减速箱低速轴中心线的交角应小于10'
3. 装配时，轴承、齿形联轴器应加适量润滑脂。
4. 零部件装配完成后，定位采用楔形挡块，楔形挡块应有效固定；定位面有效接触，敲击力应不大于50kg，内侧定位块点焊。
5. 零部件装配完成后，定位采用螺栓挡块，螺栓挡块应有效固定：螺栓头与定位面有效接触，预紧 力应不大于50kg，锁紧螺母拧紧。
6. 主要螺栓的预紧力见下表.
7. 吊具到达最低起升高度时卷筒上至少应保留压绳圈2圈、安全圈3圈。吊具到达最大起升高度时留1 个空圈。
8. 联轴节、制动器等外购零部件安装时应遵守相关零部件使用说明书中的安装要求。
9. 卷筒制动盘、低速联轴节与卷筒螺栓法兰接触面装配前只允许涂环氧富锌底漆并须保持清洁、干燥、 无油污等污染物。

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
PROJECT: Sri lanka CICT 12 STS Crane		DESIGN STAGE:			SCALE: 1:20		
ITEM NAME: 主起升机构 Main Hoist Device		DRAW NO: J526A0300			PROJ.		
ZPMC	DSGN		TRACE		APP.	Ass.	
	DRAW		CHCK		SET/CRANE	1	
	CHCK		VER.		WEIGHT	29361.4 kg	

NO.	SUB-NO. or STDD		NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
B6	GB97.1-85		平垫圈 24 Washer	300HV	104	0.05	5.2	Dacro
B5	ZN6175-T-SL		凸缘螺母 M24 Nut	10	52	0.2	10.4	Dacro
B4	GB5782-86		螺栓 M24×120 Bolt	10.9s	52	0.5	26.0	Dacro
B3	GB97.1-85		平垫圈 30 Washer	300HV	48	0.1	4.8	Dacro
B2	ZN6175-T-SL		凸缘螺母 M30 Nut	10	24	0.3	7.2	Dacro
B1	GB5782-86		螺栓 M30×120 Bolt	10.9s	24	1.0	24.0	Dacro
17	J526A0312		防跳绳装置 Rope protection	部件 Ass.	2	60.2	120.4	
16	AS1-48		调整垫片48 shims	stainless steel	10	0.0	0.0	具体数量及厚度现场定
15	ZT2-0051C(e1)		调整垫片 shims	stainless steel	8	0.0	0.0	具体数量及厚度现场定
14	ZPDR08-1		楔形块 wedge block	35	8	1.0	8.0	Standard
13	ZPDR08-2		楔形挡块 wedge block	35	4	4.9	19.6	Standard
12	RCA08		钢丝绳压板组件8 Rope clamp 8	组件 Ass.	24	3.4	81.6	Standard
11	J526A0311		卷筒轴承座 Bearing pedestal	部件 Ass.	2	409.2	818.4	三表另列
10	J526A0310		集油盘 2 Drip pan2	Q235	2	1.1	2.2	
9	J526A0309		罩壳2 cover 2	焊接件 Wel.	1	13.5	13.5	
8	J526A0308		罩壳1 cover 1	焊接件 Wel.	2	32.4	64.8	
7	J526A0307		限位装置 limit device	组件 Ass.	1	124.1	124.1	三表另列
6	J526A0306		轴端挡板2 Shaft end plate 2	Q235	2	14.3	28.6	
5	J526A0305		挡板1 shaft end plate 1	Q235	1	2.3	2.3	
4	J526A0304		集油盘 Drip pan	Q235	4	29.9	119.6	
3	J526A0303		制动盘 drum brake disc	A709-50-2	2	921.3	1842.6	
2	J526A0302		卷筒 Drum	焊接件 Wel.	2	3263.3	6526.6	
1	J526A0301		底架 Base frame	焊接件 Wel.	1	1078.9	1078.9	

NO.	SUB-NO. or STDD		NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
B28	GB97.1-85		平垫圈 30 Washer	300HV	8	0.1	0.8	发黑处理
B27	ZN6175-T-SL		凸缘螺母 M30 Nut	10	4	0.3	1.2	发黑处理
B26	GB27-88		铰制螺栓 M30×120×m6 Bolt	10.9s	4	1.0	4.0	发黑处理
B25			联轴器 AFN SB160-900x30 Coupling	外购成品 Pur.	1	290.0	290.0	"KTR"
B24			联轴器 AFN SB160-900x30 Coupling	外购成品 Pur.	1	250.0	250.0	"KTR"
B23			减速器 ML3PSF130E-1600 Speed reducer	外购成品 Pur.	1	8800.0	8800.0	"SEW"
B22			制动器 SB315 Arrester	外购成品 Pur.	2	860.0	1720.0	丰城自带紧固件及支架(详见订货图)
B21			电机 BDK-1KMU Motor	外购成品 Pur.	2	2800.0	5600.0	"YASKAWA"
B20			制动器 YP41-Ed4500-80-900x30 Brake	外购成品 Pur.	2	410.0	820.0	"ZPMC"
B19			低速联轴器 DC08B Coupling	外购成品 Pur.	2	440.0	880.0	"ZPMC"
B18	GB32.1-88		螺栓 M30×60 Bolt	8.8	4	0.4	2.4	Dacro
B17			钢丝 φ2 wrie	不锈钢 sta.stl	1	0.0	0.0	~1 m
B16	GB32.1-88		螺栓 M16×35 Bolt	8.8级	4	0.1	0.4	Dacro
B15	GB97.1-85		平垫圈 30 Washer	200HV	8	0.1	0.8	Dacro
B14	ZN6175-T-SL		凸缘螺母 M30 Nut	8	8	0.3	2.4	Dacro
B13	GB5782-86		螺栓 M30×120 Bolt	8.8	8	1.8	14.4	Dacro
B12	GB97.1-85		平垫圈 24 Washer	200HV	8	0.05	0.6	Dacro
B11	ZN6175-T-SL		凸缘螺母 M24 Nut	8	8	0.1	1.2	Dacro
B10	GB5782-86		螺栓 M24×95 Bolt	8.8级	8	0.4	4.8	Dacro
B9	GB97.1-85		平垫圈 48 Washer	200HV	10	0.4	2.4	Dacro
B8	ZN6175-T-SL		凸缘螺母 M48 Nut	8	10	1.2	7.2	Dacro
B7	GB5782-86		螺栓 M48×230 Bolt	8.8级	10	5.0	30.0	Dacro

PROJECT: Sri lanka CICT 12 STS Crane

DESIGN STAGE:

SCALE: 1:20

ITEM NAME: 主起升机构 Main Hoist Device

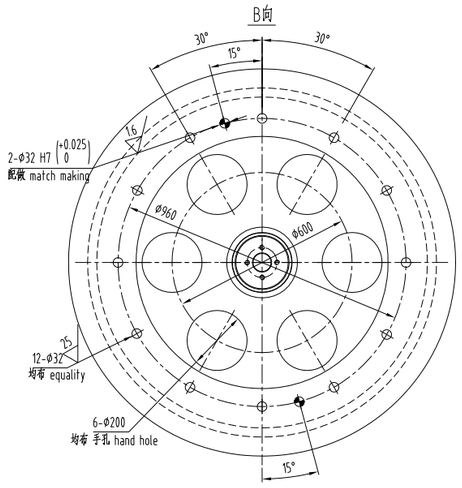
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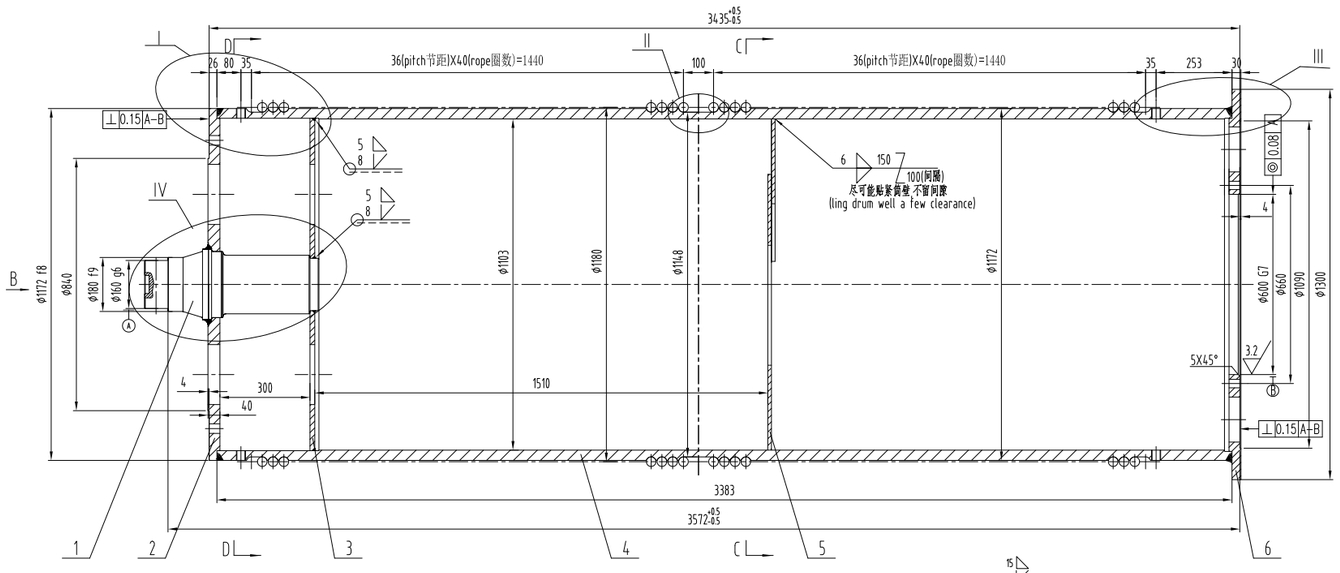
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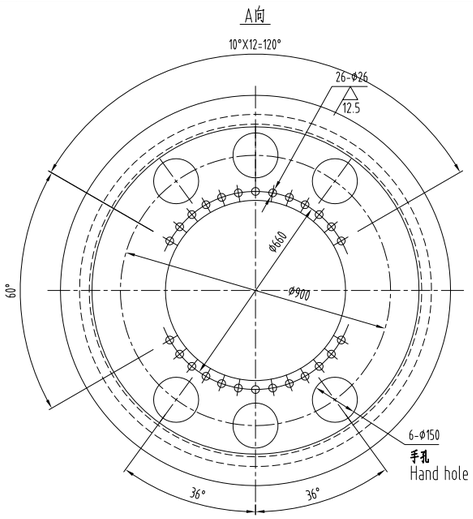
ZPMC

DSGN		TRACE		APP.	Ass.
DRAW		CHCK		SET/CRANE	1
CHCK		VER.		WEIGHT	29361.4 kg

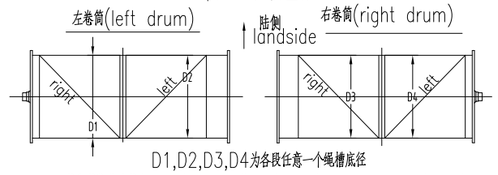
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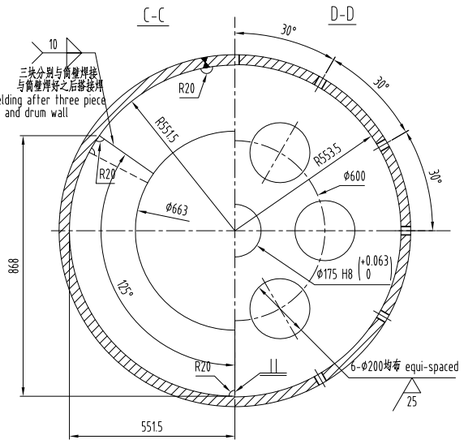


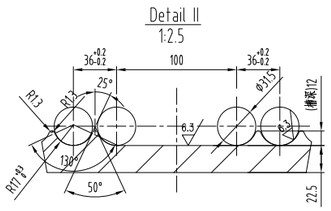


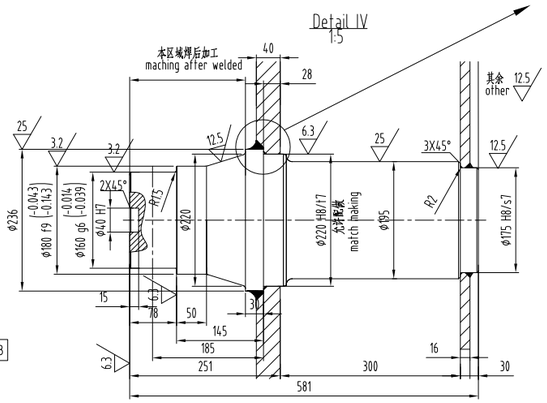
|D1−D4|<±0.1
|D1−D3|<±0.1
|D2−D3|<±0.1
|D2−D4|<±0.1

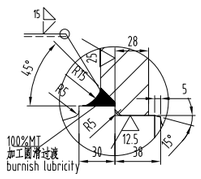


D1,D2,D3,D4为各段任意一个绳槽底径
D1,D2,D3,D4 are bottom diameters of rope groove of random







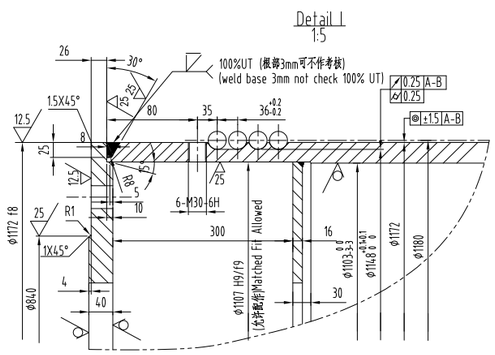


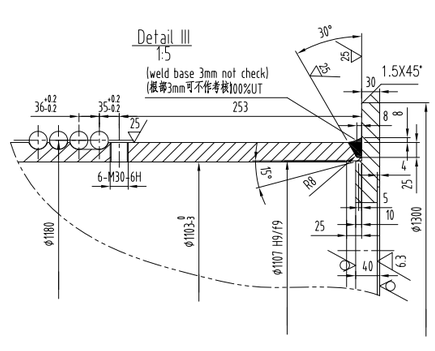
卷筒 技术要求

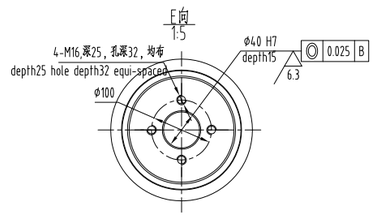
1. 卷筒分为左，右卷筒，对称制作，详见附图
2. 卷筒壁用Q345B钢板冷卷焊接制成。
3. 卷筒壁横、纵向拼接应按图纸要求开坡口对接，如果采用分段制造环焊对接时，相邻分段的纵向焊缝应错开角度不小于90°，所有对接焊缝100%UT探伤检验，焊缝质量按AWS标准检查验收。
4. 卷筒壁与端板按图中要求分别作UT和MT检验，端板与轴之间的焊缝均进行100%MT探伤检验，不得有任何方向裂纹。
5. 卷筒焊后做消应力处理。
6. 机加工后装上压板及螺栓作静平衡试验，不平衡等级G6.3，允许偏心距e=0.8mm，卷筒壁上的允许不平衡质量M=2Ge/Do=4.8(kg)，其中G-卷筒重量(kg)，Do-贴平衡重处筒壁直径(mm)，平衡重沿卷筒长度方向对称分布，材料为Q235，角焊缝固定。
7. 棱边倒钝。
8. 卷筒轴为35钢，正火处理，焊前进行化学成分、硬度(HB=149~187)及超声波探伤检验，达到厂标要求后方可装焊。
9. 卷筒焊接消应力后校圆，圆度应小于等于3mm。
10.对于起升双卷筒应保证两卷筒的各段绳槽底径相对误差不大于±0.1mm(淬火前).
11.除轴伸配合处外，卷筒内、外部均按公司标准涂油漆。

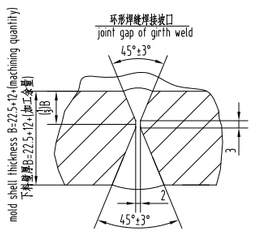
Technical requirements

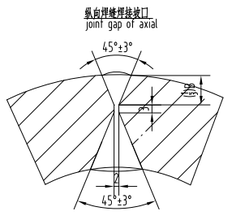
1. There are one right drum and one left drum that shall be made in symmetrically.
2. The drum shell is made by cold rolling and welding.
3. The groove of transverse and longitudinal butt weld on drum shell shall be made according to the sketch as shown in this drawing. The longitudinal butt weld between two subsections shall be shifted at least 90°in case the drum shell is fabricated in sections and connected by annular butt weld. All the butt weld shall be inspected by 100%UT. All the weld shall be made and inspected according to AWS.
4. Welds between shell and end plate shall be inspected by UT and MT as instructed in the drawing. Weld between end plate and shaft shall be inspected by MT as instructed in the drawing. Cracks in any direction are not permitted.
5. The drawing shall be vibratory stress relieved after welding.
6. The drum shall be static balanced to the grade G6.3 with rope clips and fasteners after machining. The Max allowable eccentricity e=0.8mm. The Max allowable mass on drum shell M=2Ge/Do(kg). Where: G-drum mass, Do=Distance between balance weight to CL of drum. The balance weight's material is Q235 and it shall be distributed in symmetrically in longitudinal direction and welded by filleted weld on drum shell.
7. Blunt the edges.
8. The material of shaft if 35# treated by normalizing. Inspection of Elemental compositon, hardness, and UT shall be taken before installing and welding.
9. The roundness of drum after stress relieving shall not more than 3mm.
10.For the drums used for dual hoisting crane, the deviation of bottom diameter of rope groove between right and left drums shall less than ±0.1mm before heat treating .
11. Except the fitting surface of shaft, exterior and interior surface of the drum shall be painted according to respect provisions of company.
12.The surface of rope groove will be hardened to HB300~380.



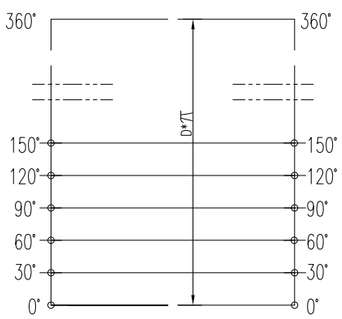








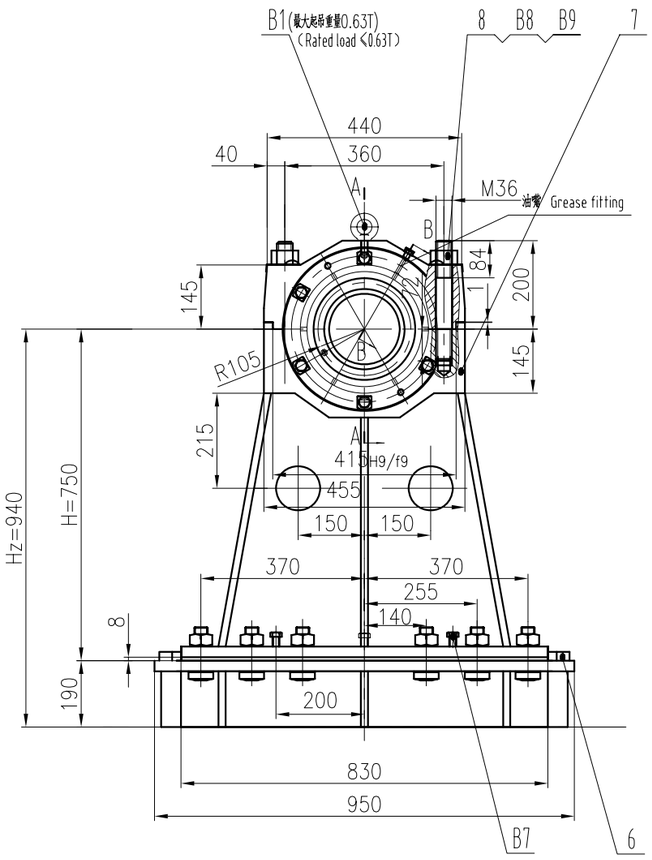
卷筒绳尾螺孔分布展开图
rope rump screws distributing spread

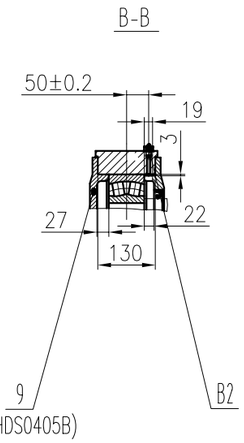


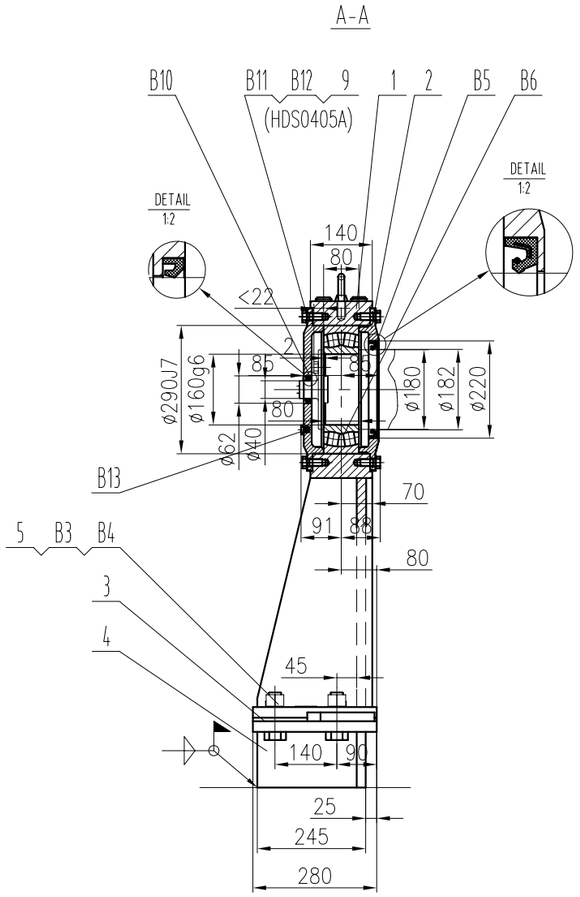
NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH(Kg)	TOTAL(Kg)	NOTE
6			钢板 40 F=φ1300/φ600 Armor plate 40	A709-50-2	1	239.8	239.8	
5			钢板 12 F=φ1103/φ663 Armor plate 12	Q345B	3	19.8	59.4	
4			钢板 (35) F=3682×3383 Armor plate	Q345B	1	2505.5	2505.5	
3			钢板 16 F=φ1107/φ175 Armor plate 16	Q345B	1	93.1	93.1	
2			钢板 40 F=φ1172/φ220 Armor plate 40	A709-50-2	1	234.5	234.5	
1			圆钢 236 L=581 Round steel 240	35	1	131.0	131.0	

设计/DSGN	wuliang	校对/CHKD		图幅/FORMAT	A1
审核/DATE	2012-6-1	审批/DATE			
材料/MAT	焊接件Wel.	比例/SCALE	1:10	图名/DRAWING NAME	卷筒 Drum
重量/WEI	3263.3				
更改标志		图号/DRAWING NO.	J526A0302	版本/REV	01
	上海振华港机公司			页号/PAGE	1

REV. No.	DESCRIPTION	DATE	signature
01		2012.06.27	W.L







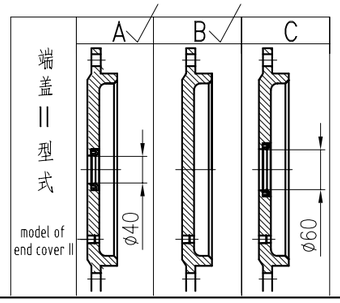
技术要求

1. 定位块ZP2现场烧焊.
2. 螺栓M16的预紧力矩为150Nm,M24为690Nm,M36为1800Nm.
3. 调整:通过调整件3"调节板"的厚度来调整中心高H,从而保证卷筒的另一端联轴节的对中性要求,调整垫片最多垫二层.
4. 加工轴承外园孔及端盖孔时,止口剖分面加1mm的工艺垫.
5. 装配时,端盖II上的注油孔布置在下部侧面,可作为放油孔.
6. 骨架密封的方向必须如图示安装.
7. 吊环螺钉(B1)仅用于起吊此部件.

Technical Requirement

1. Piece 6 (ZP2) should be welded after alignment at site.
2. Pretightening torque of bolt: M16 is 150Nm,M24 is 690Nm,M36 is 1800Nm.
3. Adjustment procedure:change height of H by using different thickness item 3 for alignment. Item 3 could not exceeded 2 piece.
4. Add 1mm washer on the split surface step of pedestal when machining bearing and hole of end cover.
5. Item B13 should be the place shown on the drawing,and it can be oil outlet
6. Item B10 (Oil seals) should be installed according to the drawing.
7. The item B1(screw) is used to lift this Bearing Pedestal only.

端盖II选型表
End Cover II Select list



油嘴选型表
Grease Fitting Select list

油嘴型号 Grease Fitting type	选择代号 choose No.		材质 material
PT1/4"	1	I	黄铜 brass
M10X1	2		
NPT1/4"	3	II	不锈钢 sta.steel
NPT1/8"	4		
	5		

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
B13	TP1	bolt螺塞 R1/4"	Q235	1			
B12	GB854-88	垫圈16	Q235	12			
B11	GB5783-86	bolt螺栓 M16x35	grade 8.8	12	0.1	1.2	
B10	HG4-692-67	oil seal骨架油封PD40x62x12		1			
B9	GB854-88	Tab-washer止动垫片36	Q235-A	4			
B8	GB6170-86	nut螺母 M36	grade 8	4	0.3	1.2	
B7	GB5783-86	bolt螺栓 M16x45	grade 8.8	3	0.1	0.3	
B6	SKF	bearing轴承 22232 CC/W33		1		24.6	ø290×ø160×80
B5	HG4-692-67	oil seal油封PD180x220x18		1			
B4	GB6170-86	nut螺母 M30	grade 8	12	0.2	2.4	
B3	GB5782-86	bolt螺栓 M30x100	grade 8.8	12	0.9	10.8	
B2		oil nozzle油嘴button PT1/4"	不锈钢 sta.stl	1			
B1	GB825-88	screw吊环螺钉 M16	20	1		0.2	

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
9	HDS0405B	end cover闷盖	Q235	1		15.3	
9	HDS0405A	end cover端盖II	Q235	1		14.9	
8	HDS0404(GB900-88)	stud 螺柱 M36x220	45(grade8.8)	4	1.4	5.6	热浸锌Hot-dip.gal
7	HDS0403-Hz=900	down pedestal下轴承座	Welded	1		174.9	
6	ZP2	locating block2号定位块	Q235	2	1.1	2.2	
5	HDS0205	washer垫片	Q235	6	0.2	1.2	借用borrow
4	J526A031101	support底座	Q235	1		79.0	借用borrow
3	HDS0203	adjusting plate调整垫	Q235	1		14.2	借用borrow
2	HDS0402	end cover端盖1	Q235	1		9.8	
1	HDS0401	upper pedestal上轴承座	Q235	1		41.4	

J526A0311	Bearing Pedestal卷筒轴承座组件		2	2x409.2	1:10

ZPMC	DRAW			TECH.			TRACE		
	CHCK			STDD			CHCK		